

BRAVO Lab

Brazilian Risk, Allocation, Volatility & Options Lab

Institutional Decision Report

PRIVATE APPLICATION DRAFT - v0.2.4

A clean decision-support report for Brazilian market stress, volatility transmission, synthetic option-overlay logic, BSTI policy selection, and portfolio-governance review.

CREATED BY

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Research, portfolio risk, derivatives overlays, and market transmission analysis

Not investment advice

EXECUTIVE DECISION MEMO

The report must answer what the portfolio should consider doing now.

BRAVO Lab currently reads Brazilian risk through a BSTI score of ****14.13****, classified as ****Calm****, with ****Brazil Drawdown Pressure**** as the dominant pressure channel. The current BSTI policy action is ****Passive Brazil Equity****.

The result is not a buy or sell recommendation. It is a decision-support signal. The point is to decide whether the current state deserves monitoring only, hedge discussion, income capture through covered calls, downside protection through collars, or no portfolio action because the signal is not persistent enough.

The current read is important because it transforms the project from a collection of charts into a committee question: does the portfolio accept the upside sold by the covered-call action, or should drawdown control receive more weight because the collar remains the strongest drawdown profile?

Decision metric	Current project read
Current BSTI score	14.13
Current BSTI regime	Calm
Dominant pressure channel	Brazil Drawdown Pressure
Current policy action	Passive Brazil Equity
Dominant historical policy choice	Passive Brazil Equity (61.59%)
BSTI policy annualized active return	5.45%
BSTI policy tracking error	10.18%
BSTI policy information ratio	0.54
BSTI policy max drawdown	-25.01%

DECISION INTERPRETATION

The model is useful only if the results change the conversation.

The current policy action points toward covered-call income capture. That does not mean the portfolio should mechanically sell upside. It means the committee should ask whether the premium harvested is worth the recovery risk under the current stress state.

The best drawdown profile being associated with the collar changes the discussion. It says that if the committee's priority is capital protection, the income-first action should be challenged against the cost and benefit of downside protection.

The BSTI policy overlay having the strongest information-ratio read is not a final proof. It is a reason to inspect whether the active return survives transaction costs, tax assumptions, liquidity, roll mechanics, and

real B3 option-chain validation.

Committee question	Decision meaning
Is the signal persistent?	A fragile state can be temporary. Action should depend on persistence, not only on one print.
Is covered call acceptable?	Covered calls collect income but sell part of the recovery. This matters if the market rebounds.
Is collar protection justified?	The collar may improve drawdown control, but protection cost must be visible.
Does the policy beat passive after friction?	The active-risk result matters only after costs, liquidity, taxes, and roll assumptions.
Should action be activated or monitored?	The model should support escalation discipline, not automatic trading.

EVIDENCE STACK

What has actually been built in the BRAVO project.

The truthful version of BRAVO is evidence-first. It combines market data, derived risk metrics, regime classification, synthetic option overlays, active-risk diagnostics, multi-asset stress signals, BSTI validation, policy selection, and implementation warnings.

The report should not bury that structure under style. The correct design is clean, but the content must remain concrete: what is observed, what is modeled, what is synthetic, what is validated, and what is still not ready for production.

Layer	What it contributes to decision-making
Market state	Tracks Brazilian equity exposure, external Brazil exposure, global equity risk, USD/BRL, and VIX.
Regime diagnosis	Classifies volatility and drawdown state before deciding whether overlay behavior is justified.
Synthetic overlay engine	Compares passive equity, covered call, collar, and stress-aware switching under controlled assumptions.
Active-risk diagnostics	Tests whether deviation from passive equity is rewarded by information ratio, hit rate, and downside behavior.
BSTI	Converts multi-asset pressure into a monitorable Brazil Stress Transmission Index.
Policy selection	Maps stress levels into passive exposure, covered-call income, or collar protection.
Limits	Separates real data from synthetic assumptions before production use.

MODEL TRUTH

The report must separate observed data, modeled signals, and synthetic assumptions.

A serious decision report cannot treat every number as equally real. The market layer uses public market price series. Realized volatility and drawdown are derived from those series. The regime classifier and BSTI are model-generated signals. The covered-call and collar overlays are synthetic until real B3 option chains are integrated.

This distinction protects the project. Real price series can support market-risk measurement. Synthetic option premiums can support research design and decision logic. They cannot yet support live execution claims without observed strikes, maturities, bid-ask spreads, liquidity filters, tax treatment, and roll rules.

Evidence type	Decision status
Public market prices	Usable as market proxies for risk measurement and historical diagnostics.
Derived metrics	Realized volatility, drawdown, VaR, CVaR, and active-risk metrics are model-derived from observed prices.
Regime classifier	Decision signal, not an observed market fact. It must remain auditable.
Synthetic covered call	Research assumption until real B3 option-chain evidence is added.
Synthetic collar	Research assumption for protection logic, not a live tradability claim.
BSTI policy rule	Governance rule requiring validation before portfolio activation.

BASELINE RISK AND OVERLAY LOGIC

The first decision is whether the portfolio should stay passive, sell volatility, or protect downside.

The baseline risk layer gives the context for the overlay discussion. Brazilian equity exposure carries meaningful drawdown risk. FX, external Brazil exposure, global equity, and VIX provide additional pressure channels. The overlay engine then asks how passive exposure compares with covered-call income, collar protection, and stress-aware switching.

The correct interpretation is not that one line wins forever. Passive exposure keeps upside but accepts drawdown. Covered calls reduce volatility and collect premium but cap recovery. Collars improve drawdown control but carry protection cost. Stress-aware switching is promising only if the signal is robust enough to justify action after friction.

Strategy	Decision read
Passive Brazilian equity	Default exposure. Strongest when participation matters and the stress signal is not persistent enough.
Covered call	Income-capture response. Useful in fragile/range-bound conditions, dangerous if it sells recovery too cheaply.
Collar	Protection response. Useful when drawdown control matters more than full participation.
Stress-aware overlay	Governance response. Useful only if regime classification survives validation and costs.

Cumulative performance path

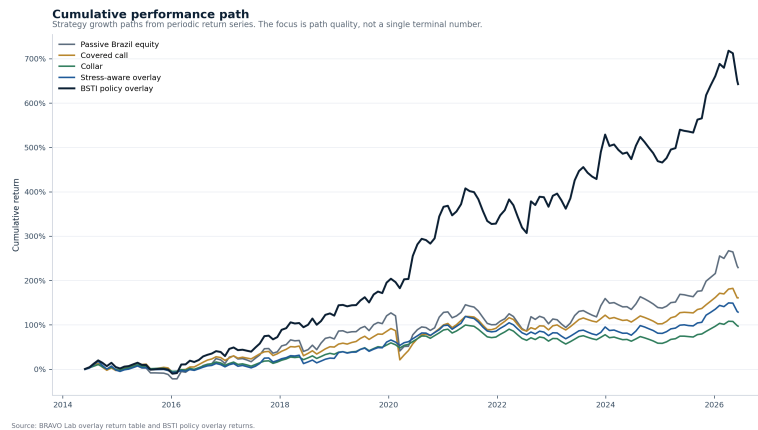


Figure: 01_cumulative_performance.png

This chart should not be read as a decorative return race. It asks whether the overlay logic changes the investment path enough to justify its active risk, implementation burden, and potential sacrifice of upside.

Drawdown control map

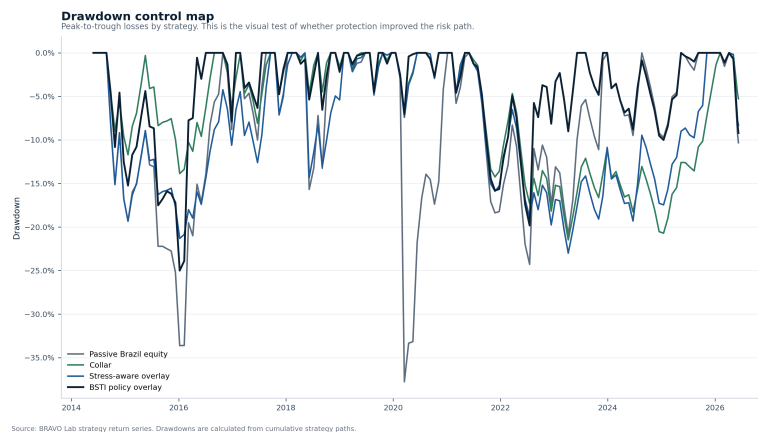


Figure: 02_drawdown_profile.png

This is one of the most decision-relevant charts in the report. If drawdown control is the priority, the collar deserves attention even if it is not the highest-return path. If income is the priority, covered calls must be judged against the risk of selling recovery.

ACTIVE-RISK DECISION LAYER

The overlay is not useful unless its deviation from passive exposure is governable.

Absolute return is not enough. A portfolio committee needs to know whether the overlay earns enough to justify tracking error against passive Brazilian equity. This is why the project includes active return, tracking error, information ratio, hit rate, downside hit rate, and worst active period.

The regime dimension is even more important. An overlay can look acceptable on average and still be dangerous in a specific market state. BRAVO's active-risk-by-regime logic tests whether covered calls,

collars, and stress-aware switching remain governable in calm, fragile, stress, and extreme-stress conditions.

This is where the project becomes decision-oriented. The purpose is not to maximize one summary statistic. The purpose is to know when a strategy helps, when it hurts, and when the committee should refuse to act because the active risk is not justified.

Active-risk question	Why it matters
Tracking error	Measures how far the overlay moves away from passive Brazil equity.
Information ratio	Tests whether active deviation is rewarded.
Hit rate	Shows how often the overlay beats passive.
Downside hit rate	Tests whether the overlay helps when passive exposure is already losing money.
Worst active period	Shows the cost of being wrong at the wrong time.

MULTI-ASSET STRESS AND BSTI

The stress signal must explain where the pressure is coming from.

BRAVO is stronger when it does not rely only on local Brazilian equity behavior. The multi-asset stress layer brings in external Brazil exposure, global equity pressure, USD/BRL pressure, and VIX pressure. This helps the committee distinguish local drawdown from broader transmission pressure.

The Brazil Stress Transmission Index then compresses those pressure channels into a 0 to 100 monitorable state while preserving the dominant channels behind the score. The purpose is not to forecast returns. The purpose is to give the portfolio committee a common language for monitoring stress, warning escalation, and overlay discussion.

BSTI output	Decision meaning
Score	Measures the intensity of current Brazil stress transmission.
Regime	Classifies the state as calm, fragile, stress, or extreme stress.
Dominant channel	Identifies whether pressure is coming from drawdown, FX, external Brazil exposure, global risk, or VIX.
Stress breadth	Shows whether pressure is narrow or spread across channels.
Policy action	Turns the signal into passive, income, or protection review.

Brazil Stress Transmission Index

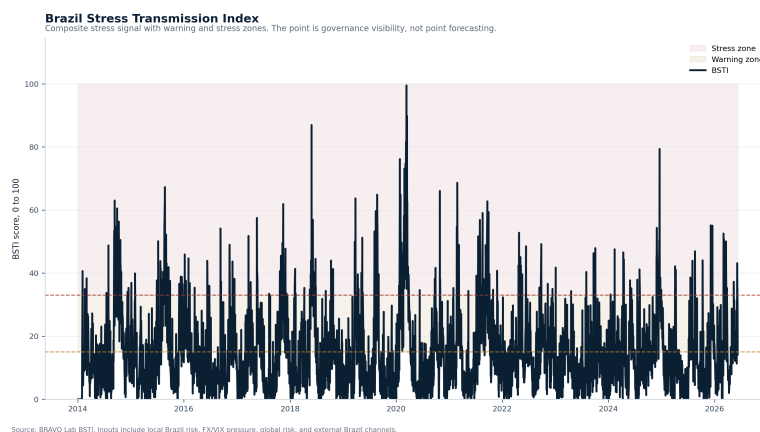


Figure: 03_bsti_signal.png

The BSTI chart makes warning and stress zones visible through time. It should be read as a governance signal: when the index rises, the committee should review whether passive exposure, covered-call income, or collar protection is more defensible.

VALIDATION AND CALIBRATION

A stress index matters only if thresholds connect to portfolio outcomes.

The validation layer is central to the project's credibility. A stress index should not exist merely because it looks sophisticated. It must be tested against future benchmark losses, drawdowns, and overlay behavior when stress is elevated.

The calibration layer asks whether alternative thresholds and channel weights create more useful governance signals. The goal is not to overfit the index. The goal is to understand whether the chosen structure is robust enough to support committee discussion.

This is why the report should not hide uncertainty. If a threshold shows stronger drawdown-event precision but weak recall, that is a decision trade-off. If a broader threshold catches more stress episodes but creates more false alarms, that is also a governance trade-off.

Validation result type	Decision implication
5 percent drawdown precision	Tests whether a high BSTI state correctly flags drawdown risk.
10 percent drawdown precision	Tests whether severe loss events are better captured by elevated stress thresholds.
Recall	Tests whether the model misses too many future stress events.
Governance score	Balances precision, recall, and usefulness for decision review.
Policy behavior under elevated BSTI	Tests whether covered calls, collars, or switching add value when stress is already high.

BSTI calibration scorecard

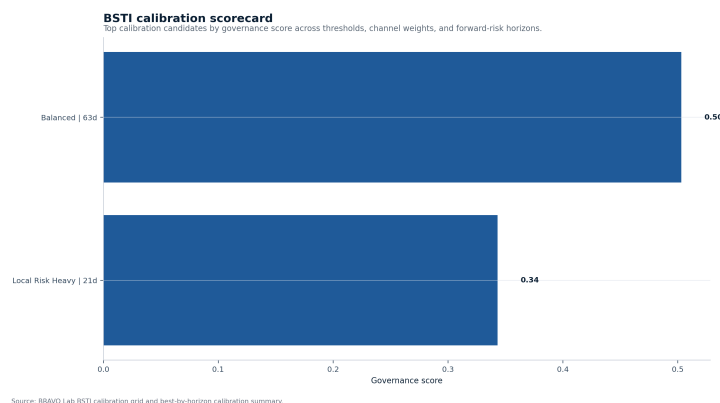


Figure: 07_bsti_calibration_scores.png

The calibration scorecard should be read as an audit tool, not as final proof. It helps identify which threshold and weighting combinations create a more useful warning signal for portfolio governance.

POLICY SELECTION AND PERSISTENCE

The rule must discipline action, not create trading noise.

The BSTI policy layer converts stress states into portfolio actions: passive exposure when stress is low, covered-call income when stress is moderate, and collar protection when stress is high. This is simple by design because a decision rule must be understandable before it becomes more complex.

Persistence matters because one warning print should not automatically trigger action. A state that remains fragile or escalates deserves more attention than a random flash. That is why the transition matrix and warning-to-stress escalation rate belong inside the decision report.

The result should be used as an agenda for review: confirm persistence, compare covered call against collar, inspect active risk, and only then decide whether the portfolio needs monitoring, hedge discussion, or actual overlay activation.

Policy condition	Suggested committee response
Low BSTI	Stay passive unless other risk controls are triggered.
Fragile BSTI	Review covered-call income capture and whether upside sold is acceptable.
Stress BSTI	Review collar protection, drawdown tolerance, and implementation cost.
Persistent warning	Escalate from monitoring to hedge discussion.
Weak or temporary signal	Avoid unnecessary action and preserve optionality.

BSTI policy-selection mix

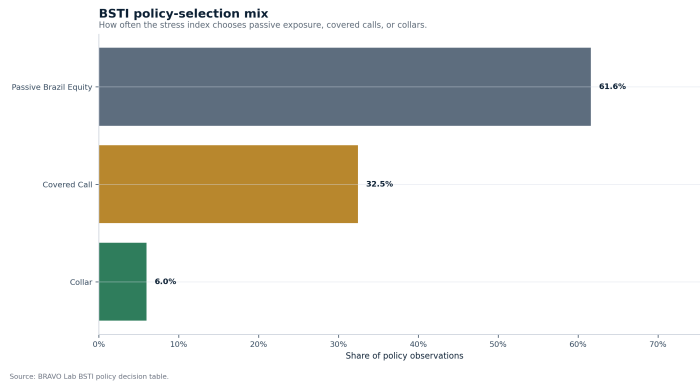


Figure: 04_bsti_policy_selection_mix.png

The policy mix shows how often the system chooses passive exposure, covered-call income, or collar protection. A useful policy should not switch constantly, but it should change behavior when the evidence deserves it.

BSTI state-transition matrix

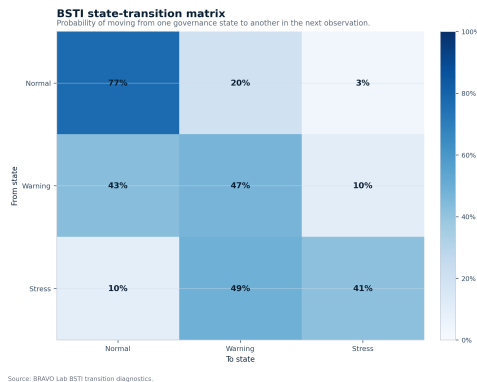


Figure: 06_bsti_transition_matrix.png

The transition matrix helps separate persistent stress from random noise. This is essential because the cost of unnecessary trading can be as damaging as the cost of ignoring real stress.

IMPLEMENTATION LIMITS

The project is decision support, not production trading infrastructure yet.

The current project is already useful as a research and application artifact, but it should not be oversold. The option-overlay layer is synthetic. The covered-call and collar premiums are research assumptions. Real B3 option-chain data must be added before the project can support live tradability claims.

The next credibility layer must include observed strikes, maturities, bid-ask spreads, liquidity filters, transaction costs, taxes, roll rules, exercise logic, and portfolio mandate constraints. Without that, the correct position is decision-support research, not investment advice.

This limitation strengthens the report if stated clearly. It tells the reviewer that the project is honest about what is built, what is simulated, what is validated, and what still needs to be improved.

Next build layer	Why it matters for decision quality
Real B3 option chains	Turns synthetic overlay research into market-observed implementation analysis.
Transaction costs and taxes	Tests whether policy benefits survive real-world friction.
Liquidity filters	Prevents the strategy from relying on options that cannot be traded efficiently.
MTV-GARCH or DCC	Improves volatility and covariance understanding.
CCA or Ridge CCA	Identifies which market blocks transmit stress most clearly.
Wavelet layer	Separates short-term noise from persistent multiscale stress.

FINAL DECISION READ

The value of BRAVO is not the chart. The value is the decision discipline.

BRAVO Lab should be presented as a reproducible Python research prototype that converts Brazilian market stress into portfolio-governance evidence. Its strongest point is not that it predicts the market. Its strongest point is that it makes the portfolio decision more explicit.

The current results support a disciplined committee workflow: read the BSTI state, identify the dominant pressure channel, inspect whether the policy action is income-oriented or protection-oriented, compare the overlay against passive exposure, review active risk, check persistence, and respect implementation limits before acting.

That is the truthful project. It is not a decorative report. It is a decision-support system in prototype form.

Evidence-first. Decision-oriented. Not investment advice.